

Administered Menus and Focal Anchors: The Microstructure of the Market for Machine Intelligence

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Abstract

Open-weight models make it possible for many firms to sell inference under the same model label. Yet the resulting market is not a conventional spot market: providers post sticky token-price menus, a router privately clears each request, and displayed quotes can be refused when selected. We construct a purpose-built panel of provider-model quotes at five-minute frequency, router-published operational aggregates, historical quote records, and randomized one-token routing probes. Three facts characterize this market. First, prices are administered menus. Only 2.8% of provider-model-days reprice, changes are large, 93% of quotes lie on cent grids, and state-dependent price gaps predict changes out of sample while congestion does not. Second, prices coordinate on a focal anchor. The two cheapest quotes tie on 45.9% of multi-provider model-days, versus 13.4% in an independent grid-constrained null. Where the model author operates an endpoint, 65 of 72 tied minima equal the author’s first-party price. Third, the router manufactures firmness from revocable quotes. In a randomized-crossover release, default routing succeeds on 151 of 152 requests, while three pinned single-provider policies succeed on 61–64 of 76; rejection is flat in price rank, refuting the stale-cheap last-look pattern suggested by an earlier fixed-order pilot. A steering audit provides the mechanism link: conditional on currently being cheapest, providers that cut price in the prior week receive 3.9% default selection share versus 23.3% otherwise. Thus the observed rule does not reward past undercutting in the manner proposed to discipline pricing algorithms. The combined evidence supports a dealer-and-dispatcher analogy rather than an automated-market-maker analogy: money prices are sticky, quantity and eligibility clear at request time, and the platform’s substitution and prominence rules are first-order market-design instruments.

1 Introduction

The supply of machine intelligence has separated from model ownership. A model author releases weights or an API-compatible specification, independent inference providers host the same model, and a routing platform presents those providers as substitutable sources. A user or application chooses the model label; the router then chooses a provider, retries, or falls back. This vertical separation creates a market whose economic object is not simply “which model is best?” but “how does a platform clear multiple suppliers of the same model?”

That question is difficult to answer from posted prices alone. The price surface is public, but the allocation rule and capacity state are private. The same cheap quote can be visible to every user while being unavailable to a particular request. Conversely, a router can deliver a high fill rate even if no individual quote is firm, by substituting across providers. Treating the public surface as an order book therefore confuses displayed liquidity with delivered liquidity. Treating the router as a decentralized-exchange aggregator makes a related mistake: an inference request is not atomically split across providers, the price is not an executable commitment, and selection depends on hidden eligibility and quality.

We characterize the market with a measurement design that joins four kinds of evidence: high-frequency quote surfaces, router-side operational aggregates, historical price records, and our own randomized micro-purchases. The design keeps three estimands separate. Public observations measure the displayed menu; published operational aggregates measure congestion and enforcement; and owned probes measure the realized policy for one account and workload. None alone identifies market-wide flow, private capacity, or provider cost.

The paper’s contribution is three empirical facts and one mechanism audit.

1. **Administered menus.** Prices move infrequently and by large increments. A nested discrete-time hazard rejects a constant-hazard benchmark [Calvo, 1983]: duration and calendar effects matter, and the provider’s price gap predicts repricing both in sample and on later days. Rival changes improve the in-sample fit but not yet out-of-sample discrimination; congestion variables do not improve the hazard. The money price is therefore a sticky menu, not the fast-clearing state variable.
2. **Focal anchoring.** Minimum-price ties are too common to be a rounding artifact. An independent grid-resampling null reproduces the observed price grid and heterogeneity but yields only a 13.4% tie rate, compared with 45.9% in the data. The focal level is economically interpretable: among tied models with an author-operated endpoint, 90.3% tie at the author’s first-party price. This adapts the nonbinding-focal-point design of Knittel and Stango [2003] to a platform-defined reference price while stopping short of a collusion claim.
3. **Manufactured firmness.** The money quote barely moves while operational state changes frequently. A prospective crossover randomizes the order of default routing and three pinned-provider policies within model-time blocks. Default routing succeeds at 99.3%; pinned routes succeed at 80.3–84.2%. The common rejection level survives randomization, but an earlier apparent price-rank gradient does not. Individual quotes are revocable for the studied account, and the router’s fallback option creates the delivered product.
4. **Steering memory.** Johnson et al. [2023] show that prominence rules which reward past price cuts can disrupt algorithmic collusion. Our revealed-policy audit asks whether current cheapest providers receive more selection after a recent cut. They receive less. This is not proof of collusion—eligibility remains partly hidden—but it rejects the sign of the proposed pro-competitive implementation in the observed sample.

These facts jointly imply a sharper market analogy. The model is a product standard or protocol; an inference provider is a dealer with a revocable quote and hidden inventory; the router is a dispatcher and delegated purchasing agent; and the harness is an application-side broker controlling retries and fallback. The closest mature markets are request-for-quote dealer markets and platform procurement, not constant-function automated market makers. The analogy matters because market quality depends on admission, substitution, and steering rules, not only on quoted spreads.

Our results contribute to three literatures. Work on language-model routing optimizes quality-cost choice across models Chen et al. [2023], Ong et al. [2024], Hu et al. [2024]; we instead measure competition and execution among providers of the same model. Brown and MacKay [2023] show how heterogeneous pricing technology changes competition; we provide the first high-frequency pricing-technology screen for inference providers, while making its short-panel limits explicit. Platform design work studies prominence and algorithmic pricing [Johnson et al., 2023]; our audit turns its comparative static into a statistic that a key-holder can reproduce from realized selections.

2 Institutional setting and economic objects

2.1 The transaction

A request names a model m , supplies an input, and may specify provider preferences. For provider i at time t , the displayed menu contains an input token price p_{imt}^{in} and an output token price p_{imt}^{out} . For a fixed workload with a input and b output tokens, the displayed money quote is

$$q_{imt}(a, b) = ap_{imt}^{\text{in}} + bp_{imt}^{\text{out}}. \quad (1)$$

The platform can additionally observe health, throughput, latency, context-window compatibility, account-specific limits, and contractual eligibility. Let $e_{imtu} \in \{0, 1\}$ denote eligibility for user u , and let c_{imt} denote unobserved deliverable capacity. The default router selects a provider according to an unknown score

$$i_{mtu}^* = \arg \max_i S(q_{imt}, e_{imtu}, c_{imt}, z_{imt}, h_u), \quad (2)$$

where z collects measured quality and h_u is account state. A pinned request sets the candidate set to one provider and can disable fallback. Comparing default and pinned execution therefore measures the value of delegation, not merely a price effect.

The request is served by one provider. Fallback may create multiple attempts, but this is sequential substitution rather than splitting one prompt across a portfolio. For the interfaces studied here, the route cannot generally be observed as a firm executable allocation before purchase. This absence of a public pre-trade route is why literal order-flow front-running is not identified from public data.

2.2 Measurement levels

We use an explicit observability hierarchy. It prevents a common error in this market: assigning the identifying content of realized flow to a price-only simulation.

Level	Object	Identified use
L0	Public quote surface	Price levels, gaps, ties, grids, and repricing events
L1	Documented policy simulation	Mechanical ranking or allocation proxy only
L2	Public operational aggregates	Congestion, rate-limit, derank, and enforcement state
L3	Owned realized requests	Selected provider, success, latency, cost, and fallback for the probed account
L4	Provider or router logs	Capacity commitments, full attempt order, and cross-user flow

Table 1: Observability hierarchy. The core paper uses L0–L3. No claim that requires cross-user order flow or private provider cost is made.

Public token-allocation displays sometimes expose a deterministic price-weighting proxy. When a documented inverse-square proxy is available, we compute

$$\hat{s}_{imt} = \frac{q_{imt}^{-2}}{\sum_j q_{jmt}^{-2}}, \quad \frac{\partial \log \hat{s}_{imt}}{\partial \log q_{imt}} = -2(1 - \hat{s}_{imt}). \quad (3)$$

Equation (3) is a policy arithmetic identity. It is not demand elasticity and is never substituted for realized routing in our causal claims.

3 Data and research design

3.1 Capture system

The live panel snapshots public provider-model menus every five minutes. Each record stores a UTC capture time, stable provider and model identifiers, input and output token prices, capacity ceilings when displayed, and router-published five- and thirty-minute enforcement aggregates. Captures are append-only and mirrored to remote object storage. The analysis also uses a three-year historical quote registry for lifecycle checks, public model-level token-allocation records, and owned generation records for requests made with our study key.

The pricing freeze used in the main tables contains 2,968,449 endpoint snapshots from 1,805 capture runs over nine days. It covers approximately 300 models and 69 providers with enough exposure for provider-level cadence classification. The all-field event ledger contains 3,212 changes; the Brown–MacKay completion-price screen retains 278 positive-price events spanning 7.53 days. The public routing simulation contains 562,550 provider-model-time rows over six days. Owned legacy traffic contains 2,659 raw rows over 65 runs; after deduplication and exclusion of outcome-masked studies, 716 descriptive attempts remain and 617 identify the selected provider. The randomized crossover is a separate prospective release.

Panel	Rows / attempts	Span	Primary role
Endpoint snapshots	2,968,449	9 days	Prices, grids, gaps, enforcement state
All-field price events	3,212	9 days	Repricing incidence and event construction
Completion-price events	278	7.53 days	Pricing-technology cadence screen
Public allocation proxy	562,550	6 days	Mechanical share-price comparative statics
Legacy owned attempts	716	65 runs	Descriptive provider-selection audit
Randomized crossover	380	prospective	Quote firmness and policy-order correction
Historical quote registry	multi-year	3 years	Lifecycle and duration sensitivity

Table 2: Frozen analysis panels. Sample sizes refer to the v3 manuscript release and are updated only at registered promotion gates.

3.2 Cleaning and event construction

Prices are converted to dollars per million tokens before comparisons. We use standard provider variants and remove exact duplicate captures. A price event is the first observation at which either token price differs from the preceding observation for the same provider-model pair. Five-minute timing creates interval censoring within a capture bin; we do not infer within-bin order. The first price spell in a panel is left-censored and is excluded from moments that require age.

Operational counters are router-published aggregates, not raw request logs. Their reporting can be intermittent. Regressions that use utilization or rate-limit state include a missingness indicator and are repeated on the complete-reporting subsample. Status-derived incidents are labeled as heuristics unless the source publishes exact incident timestamps. Owned probes record only metadata needed for the study-assignment, policy, provider, latency, token count, cost, and status—and never retain prompt payloads.

3.3 Claim discipline

The empirical sequence was adaptive, so we distinguish discovery from confirmation. Early descriptive analyses generated the administered-menu, focal-anchor, and firmness hypotheses. The fixed-order firmness pilot revealed a large default-pinned gap but confounded policy with position. We disclosed that failure, froze a randomized-crossover replacement before observing its outcomes, and treat the first release as interim. Dynamic pricing results are re-estimated at a registered 30-day gate using both the original and expanded vintages. Tests that have not reached their registered support are reported as power-gated rather than as null findings.

4 Fact 1: prices are administered menus

4.1 Frequency, size, and the clock

Only 2.8% of provider-model-days contain a price change. Across completed endpoint spells, the implied average duration is 26.7 days. Conditional on a change, the median absolute log change is 0.134; two thirds are cuts. Ninety-three percent of quotes lie on cent-per-million-token grids. Twenty-six percent of cuts occur at 00:00 UTC, a concentration difficult to reconcile with a continuously clearing spot price and consistent with scheduled menu updates.

Raw pooled price-change kurtosis is 113 because providers and model scales differ. After standardizing changes within provider, kurtosis is 3.54; the provider median is 3.4 with interquartile range 2.1–4.9. This statistic lies between a simple state-dependent menu-cost benchmark and the Calvo benchmark discussed by Alvarez et al. [2016]. We use it as a sufficient-statistic diagnostic, not as a structural estimate: the short panel leaves many providers with too few changes for stable within-unit moments, and observation costs can mimic infrequent adjustment Alvarez et al. [2011].

4.2 A nested repricing hazard

For provider-model pair k on day d , let $Y_{kd} = 1$ if a price changes. We fit a nested logit ladder

$$\mathbb{P}(Y_{kd} = 1 \mid X_{kd}) = \Lambda(\alpha + T_{kd} + G_{kd} + C_{kd} + R_{kd}), \quad (4)$$

where T contains age and calendar terms; G contains the absolute and signed gap from the cheapest rival plus GPU-cost changes; C contains published utilization and rate-limit state; and R contains rival raises and cuts in the prior day. The ladder adds these blocks sequentially. Provider activity is included as a frailty proxy, but with only nine days we do not estimate a full unit-frailty duration model.

The constant-hazard model is decisively rejected by duration and calendar terms. Adding price gaps improves fit ($p = 1.0 \times 10^{-5}$); adding congestion does not ($p = 0.17$); adding rival moves improves in-sample likelihood ($p = 0.0018$). Out-of-sample results are more disciplined. On a day split, time dependence alone has AUC 0.525 and the state-dependent model has AUC 0.638. The strategic model has AUC 0.555 on 36 test events. Thus the gap channel survives temporal validation; the rival-reaction channel does not yet generalize.

Hazard rung	Added state	Incremental evidence	Day-split AUC
Constant	none	baseline	0.500
Time-dependent	duration, clock	rejects constant hazard	0.525
State-dependent	own-rival gap, GPU change	LR $p = 1.0 \times 10^{-5}$	0.638
Congestion	utilization, rate limits	LR $p = 0.17$	not promoted
Strategic	rival moves	LR $p = 0.0018$ in sample	0.555

Table 3: Nested repricing hazard. The strategic rung is labeled in-sample-only; the 30-day both-vintage gate is preregistered.

The appropriate conclusion is narrower than “providers use pricing algorithms.” Prices respond to their distance from the menu’s competitive reference point and are updated on scheduled attention dates. We call them administered menus because the money quote is set periodically while the physical state clears elsewhere. Whether rival reactions reflect common costs, algorithmic competition, or coordinated behavior remains unresolved.

5 Fact 2: minimum prices coordinate on an author anchor

5.1 The tie atom and its null

For each model-day with at least two providers, order the completion quotes and define the relative second gap

$$g_{md} = \frac{q_{(2),md} - q_{(1),md}}{q_{(1),md}}. \quad (5)$$

Across 1,311 model-days, $g_{md} = 0$ in 45.9%. A zero gap is especially common in two-provider markets, but the atom persists in thicker markets. Among all non-minimum endpoint observations, 47.4% are exactly tied at the minimum; the remaining distribution has relatively little mass in very small undercuts.

Rounding can generate ties mechanically. We therefore construct an independent grid null. Within each model-day, each provider draws a log-price deviation from the pooled empirical distribution of deviations from model-day medians. Draws are independent across providers, added to the observed model-day median, and snapped to the observed cent-per-million-token grid. We preserve the observed number of providers by model-day and repeat the procedure. The null produces a 13.4% mean tie rate with 1.0 percentage-point simulation standard deviation. Snapping to a coarser dime grid increases the null to 27.6%, still well below 45.9%. Because the pooled deviation distribution retains some empirical tie mass, this null is conservative with respect to independent coordination on the same grid point.

Post-freeze robustness (1,456 model-days, 148 model clusters) retains a 45.1% tie rate. The cent-grid null is 12.9%, giving a 32.2-point excess (model-cluster bootstrap 95% interval [25.8, 37.8]); the dime-grid null is 27.2%, giving an 18.0-point excess [9.7, 25.1]. The bootstrap resamples whole models and re-simulates the null within each draw; it is a robustness refresh, not a replacement for the frozen v3 vintage.

Statistic	Estimate	Denominator
Observed tie at minimum	45.9%	1,311 model-days
Independent cent-grid null	13.4%	matched simulations
Independent dime-grid sensitivity	27.6%	matched simulations
Observed / cent-null ratio	3.4	–
Observed / dime-null ratio	1.7	–

Table 4: Minimum-price tie atom. The null preserves market thickness, observed deviation heterogeneity, and a discrete price grid, but breaks cross-provider dependence.

5.2 Identifying the focal level

The latest snapshot contains 146 tied models. For 72, the model author also operates a first-party endpoint. In 65 of those 72 cases, or 90.3%, the tied minimum equals the author’s price to machine precision. Sixty-one percent of tie levels are round to a tenth of a dollar per million tokens, but roundness is not an independent explanation: first-party author prices are themselves round. Direct provider APIs and router quotes are at exact parity in 99.6% of matched cases; we see almost no systematic direct-channel undercut.

The event directions complicate a collusion story. Of 28 isolated tie formations, 71.4% are formed by a downward move. Of 38 isolated tie breaks, 86.8% are broken by a downward move. Those directions are competitive. The evidence therefore supports an *anchor*—a salient level on which independent providers cluster— but does not identify why they anchor. The author price may signal cost, quality, or a voluntary most-favored-nation convention; it may also act as a Schelling-like coordination device in the sense of Knittel and Stango [2003].

The sharp discriminator is an author-price move. If third-party ties re-form around a new author price, the anchor is active; if they remain at the old level, the tie is a launch fossil. No author repricing occurs in the current nine-day window, so the registered event study has zero treated events. We report that support failure rather than treating a stable anchor as evidence for either model.

6 Fact 3: the router manufactures quote firmness

6.1 Price state and operational state

Over the pricing panel, only 12% of endpoints ever change price while 86% experience variation in published rate-limit state. At thirty-minute horizons, latency and enforcement variables load much more strongly on operational changes than price does. Raises tend to occur after slack rather than at congestion spikes. These facts suggest that scarcity is cleared through quantity, eligibility, and fallback while the menu remains fixed.

Displayed unavailability is not enough to establish quote revocability: public counters need not describe the study account. We therefore purchase minimal one-token completions under four policies. Default routing delegates provider choice and permits fallback. Pinned-cheapest, pinned-second, and pinned-random restrict the request to one provider selected from the contemporaneous public surface.

6.2 The failed pilot and randomized replacement

The first 96 complete blocks always ran default first. It found a 17.7 percentage- point default advantage over the cheapest pinned quote, but order and policy were perfectly confounded. Account-

level throttling, warm-up, or within-block interference could produce the entire gradient. We retain that result only as a design failure.

The replacement randomizes both model order and policy order for every block and records the assignment seed before execution. Let Z_{bp} be assignment of policy p to first position in block b , and Y_{bp} request success. Under random assignment, the first-position difference

$$\hat{\tau}_p^{(1)} = \frac{\sum_b Z_{bD} Y_{bD}}{\sum_b Z_{bD}} - \frac{\sum_b Z_{bp} Y_{bp}}{\sum_b Z_{bp}} \quad (6)$$

is design-unbiased despite arbitrary effects of the first probe on later probes. Pooled within-block contrasts are more precise but require that carryover is controlled by randomized order and is not policy-specific after conditioning on position. We therefore report both and make the first-position estimator the carryover-robust estimand at the promotion gate.

The first released accrual has 152 default attempts and 76 attempts in each pinned arm. Default succeeds on 151 of 152 requests, or 99.3%. Pinned-cheapest succeeds on 61 of 76 (80.3%), pinned-second on 62 of 76 (81.6%), and pinned-random on 64 of 76 (84.2%). Marginal Wilson intervals are shown in figure 1; repeated model-time blocks mean these intervals are descriptive, not the final randomization inference. The registered promotion gate requires 500 attempts per pinned arm, Holm-adjusted default-pinned contrasts, seed replay, and the first-position sensitivity.

The economically important result is the combination of a level and a null. The approximately 19% pinned rejection level survives randomization. The ranking gradient does not: the cheapest, second-cheapest, and random pinned endpoints have similar success. The pilot’s apparent stale-cheap “last look” was an order artifact. The surviving fact is provider-level revocability plus platform-level firmness. A visible quote is an eligibility advertisement rather than a firm commitment for this account, and the router assembles a reliable product by substituting among individually unreliable sources.

7 The steering audit

The platform-design problem is not exhausted by current-price ranking. Johnson et al. [2023] study dynamic price-directed prominence: a platform can reward a seller that previously cut price with additional exposure, changing the repeated-pricing game even after the current ranking is held fixed. We turn the sign of that comparative static into a revealed-policy audit.

For provider-model-day cell (i, m, d) , let S_{imd} be its share of selected providers among our default probes, C_{imd} indicate a price cut in the prior seven days, and R_{imd} denote its current price rank. The audit estimates

$$S_{imd} = \alpha_{md} + \beta C_{imd} + f(R_{imd}) + \varepsilon_{imd} \quad (7)$$

and reports the nonparametric cheapest-provider cells. Among 251 provider-model-days, a currently cheapest provider without a recent cut has mean selection share 23.3%; one with a recent cut has 3.9%. The rank-controlled coefficient is negative. The observed policy therefore does not implement the positive past-cut reward that disciplines patient pricing algorithms in the theory.

Eligibility is the central threat. Publicly quoted endpoints may be ineligible for our region, prompt, or account. Restricting the comparison to provider-model pairs whose eligibility is independently confirmed by pinned probes preserves the direction—30.0% without a recent cut versus 9.5% with a cut—but leaves only 11 and one cells. We therefore label the unrestricted result an audit statistic, not a causal effect of cutting price on future prominence. Reversal would require eligibility misclassification to be concentrated among recent cutters, a pattern the prospective pinned panel is designed to measure.

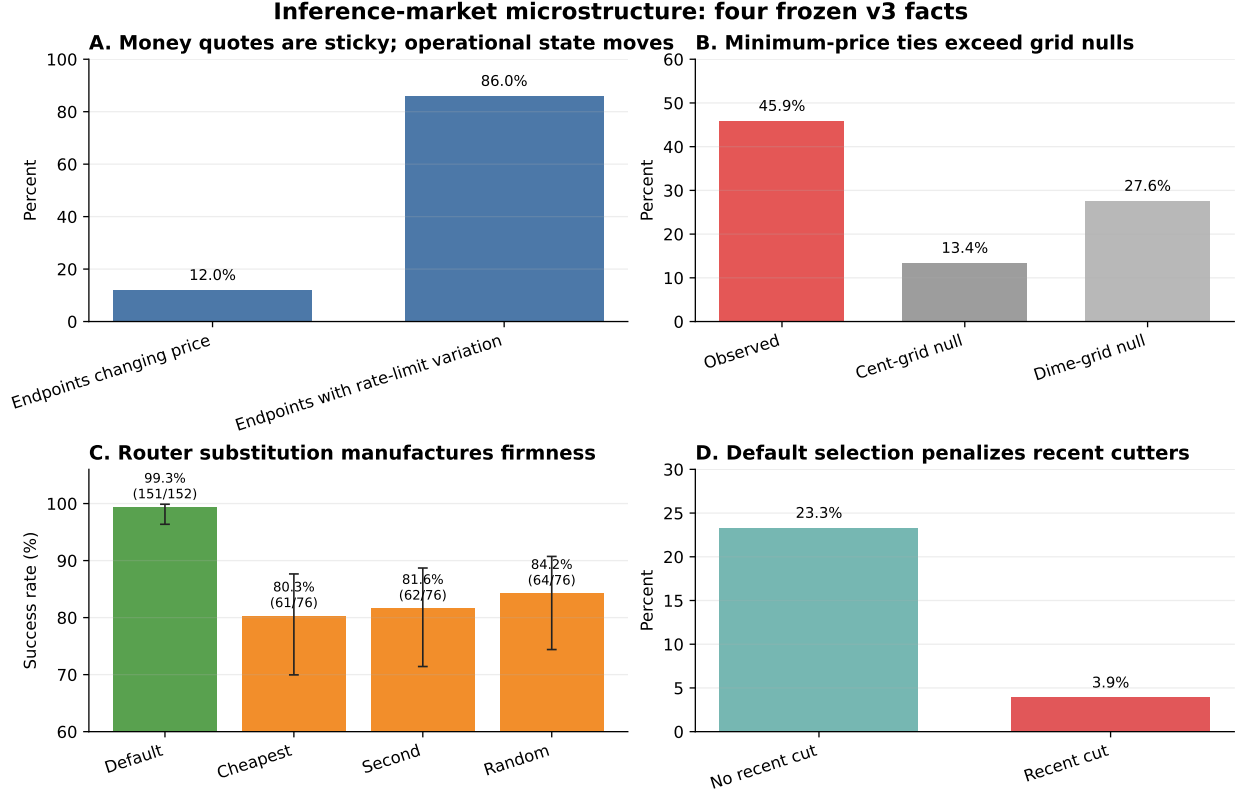


Figure 1: Four frozen v3 facts. Panel C shows marginal Wilson 95% intervals; the registered policy contrasts use block randomization inference at the 500-per-pinned-arm gate. Panels A, B, and D are descriptive panel statistics.

Two interpretations remain. Recent cuts may predict degraded capacity, causing the router rationally to protect users. Alternatively, a negative-memory rule taxes price competition. A randomized visibility test or router-side eligibility log distinguishes them. What the current data rule out is narrower but useful: for this account and sample, the revealed selection pattern does not reward past undercutting.

8 Bounded secondary evidence

This section reports results that sharpen the market description but do not carry the paper’s novelty claim.

8.1 Pricing cadence and Brown–MacKay

Of 69 exposed providers, 19 reprice at least once: ten are classified intraday, seven daily, and two weekly; 50 are inactive or left-censored. Conditional on model, slow providers quote 10.5% more than fast providers. The log-price coefficient is -0.0996 with provider-clustered standard error 0.0292 and 95% interval $[-0.1569, -0.0424]$ across 5,576 observations and 131 models.

This resembles the cadence premium in Brown and MacKay [2023], but the reaction design is not supported. Freezing cadence on the first 70% of events leaves 21 evaluation waves and 124 rival risk pairs, with zero slow-initiator/fast-responder exposures. Adding reaction features

slightly improves holdout MAE but not robust discrimination. The cadence premium is a resolved association; Brown–MacKay is a power-gated competitive null, not the selected mechanism.

8.2 Entry and long-memory demand

Across 304 models, the elasticity of quoting-provider count with respect to token demand is 0.149 (95% interval [0.126, 0.175]); the active-provider elasticity is 0.173. Both are below i.i.d. square-root and cube-root entry benchmarks. A long-memory correction maps a measured multi-scale count Hurst exponent of 0.835 to $n^{(2-2H)/2} = n^{0.165}$, numerically close to the observed slope. We treat this as a remark, not identification: entry and demand are simultaneous, the panel is short, and a deseasonalized intraday estimator is anti-persistent. The registered within-market interaction is underpowered.

8.3 Retry feedback

Within endpoint, subsequent thirty-minute request growth rises with the current rate-limit share. Demand-controlled OLS gives 0.167, while a backward placebo is -0.126 . Instrumenting focal rationing with the same provider’s rate limiting on other models gives 0.023 with interval $[-0.043, 0.085]$. The sign-asymmetric OLS is consistent with retry amplification; the instrument bounds the same-endpoint component near zero and excludes the rerouting margin. We do not convert either coefficient into welfare because request aggregates do not reveal user values, backoff policies, or cross-provider substitution.

8.4 Conduct screens

A naive event classifier labels 83% of rival responses to cuts as punish-and-revert. Reconstructing the initiator’s subsequent action changes the classification: 42.1% are punish-and-revert candidates and 41.1% are cuts the initiator later withdraws. Among raises, 65.9% are followed within 72 hours, but many initiators have low measured volume. These classifications show why posted-price event shapes alone are not proof of algorithmic collusion. Common cost shocks, experimentation, and sparse observation remain observationally equivalent without an adoption event or exogenous cost shock.

9 Market-design implications and welfare

The facts identify design wedges, not a structural welfare number. Let user type θ have completion value v_θ , delay cost d_θ , and rejection loss ℓ_θ . If provider i serves quantity x_i with capacity μ_i , quality a_i , unit cost $c_i(a_i)$, and delay $W_i(x_i, \mu_i)$, a planner chooses admission A_θ , routing $x_{i\theta}$, capacity, and quality to maximize

$$\begin{aligned} \max \quad & \int A_\theta [v_\theta(1 - r_\theta) - d_\theta W_\theta - \ell_\theta r_\theta] dF(\theta) \\ & - \sum_i [k_i(\mu_i) + c_i(a_i)x_i] - \int q_\theta(1 - a_{i(\theta)})dF(\theta), \end{aligned} \tag{8}$$

subject to capacity, queue stability, and retry feedback. Transfers cancel from global welfare but determine participation and investment.

The decentralized agents solve different problems. A provider chooses a sticky menu, capacity, and hidden quality to maximize margin times routed volume minus capacity and adjustment costs.

The router chooses admission, allocation, fallback, monitoring, and steering to maximize fees and retained user value. A harness chooses model, retry, and fallback policies to maximize application retention net of spend. A user sees the harness’s delivered price-quality-delay bundle, not the provider market directly.

The evidence isolates four possible failures of decentralization.

1. **Admission.** A high default fill rate can be privately valuable but socially excessive if retries impose congestion on other users. User-value signals are needed to price or defer low-value requests.
2. **Quality verification.** Selection on price and latency can induce unmeasured fidelity shaving. The public panel has no task-level quality certificate, so the corresponding welfare term is unidentified.
3. **Steering.** Prominence is an allocation instrument. The negative past-cut association fails the sign of a known pro-competitive dynamic rule and may reduce the return to undercutting unless it is explained by hidden quality or capacity.
4. **Retry feedback.** The OLS-placebo pattern is consistent with positive feedback, but the IV bound implies little same-endpoint amplification. The important unmeasured margin is rerouting across providers.

The central market-design implication follows directly from Fact 3. A price-only best-execution standard is incomplete when the quote is revocable. A meaningful score must combine expected money cost with success probability, latency, and fidelity for a declared workload. Capacity certificates, explicit fallback prices, or auditable provider-level fill rates would move the market from displayed menus toward executable commitments. The companion mechanism paper studies such certificates; this paper supplies the empirical reason they are needed.

10 Related work

Model routing. FrugalGPT, RouteLLM, and RouterBench study how to choose among models with different quality and cost Chen et al. [2023], Ong et al. [2024], Hu et al. [2024]. Their unit is a model. Our unit is a provider-model endpoint, holding the model label fixed and measuring the market that supplies it. The two problems compose: a harness may choose the model and a marketplace may then choose the supplier.

Pricing technology and focal points. Calvo [1983] provides the constant-hazard benchmark; observation and menu-cost models motivate duration, gap, and change-size diagnostics [Alvarez et al., 2011, 2016]. Brown and MacKay [2023] show that heterogeneous repricing frequency can soften competition. Knittel and Stango [2003] identify a nonbinding regulatory ceiling as a focal point. We import their empirical logic but use an author’s first-party price as the candidate anchor and an independent discrete-grid null as the counterfactual.

Platform steering and marketplace experiments. Johnson et al. [2023] connect prominence rules to algorithmic pricing. Our steering-memory statistic is an empirical implementation of their sign restriction. Switchback and marketplace experiments require special care because treatment changes congestion and later outcomes Bojinov et al. [2023], Holtz et al. [2025]; our first-position estimator protects the firmness contrast from arbitrary within-block carryover. Online matching under stochastic rewards provides the OR analogue for provider selection Brubach et al. [2025], but the inference setting adds sticky provider menus and hidden eligibility.

Emerging AI-platform evidence. Recent work studies quality-adjusted AI prices and platform self-preferencing Wang [2026], Zhang et al. [2026]. We complement these studies with provider-level microstructure and a reproducible realized-policy audit. The institutional APIs are documented by the platforms themselves OpenRouter [2026], Hugging Face [2026]; all policy claims in this paper are versioned to the captured interfaces rather than generalized to undocumented behavior.

11 Limitations and conclusion

Five limitations define the claim boundary. First, the main pricing panel is nine days, so dynamic hazard and cadence results await the registered 30-day both-vintage re-estimation. Second, the randomized crossover is an interim release and has not reached 500 attempts per pinned arm. Third, owned probes identify one account, region, prompt family, and time window; they do not identify market-wide provider shares. Fourth, public operational fields are router-published aggregates with missingness, not raw attempt logs. Fifth, provider cost, private rebates, task fidelity, and cross-user request order are unobserved. We therefore do not claim literal front-running, collusion, private cost, or a welfare loss.

Within those limits, the evidence changes the economic description of inference markets. Prices are not the rapidly clearing state variable. Providers post sticky administered menus, minimum quotes coordinate around the model author’s price, and the platform manufactures a firm service by substituting across revocable provider quotes. The platform’s allocation memory does not reward recent undercutting in the observed panel. These facts make the inference router a market designer: its admission, fallback, and steering rules determine the delivered product as much as the visible price surface does.

The resulting research agenda is concrete. Longer panels test whether gap-driven repricing and author anchors survive new vintages. The randomized promotion gate estimates the deliverability frontier without order confounding. Provider- or router-side eligibility logs would separate rational quality protection from a tax on undercutting. Transparent decentralized markets supply an external laboratory in which accepted bids and termination reasons are public. Together, these designs can turn a market that currently displays menus into one whose commitments, allocation, and welfare can be audited.

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A Variable construction and data-generating process

Variable	Construction
Completion quote	Output-token price in dollars per million tokens for the standard endpoint variant
Price event	First five-minute capture whose quote differs from the preceding same-endpoint capture
Cheapest rank	Dense rank of the workload-specific quote within model and capture time
Tie at minimum	At least two distinct providers have the exact minimum quote after common-unit conversion
Quote age	Time since the last observed event; first panel spell is left-censored
Recent cut	At least one negative completion-price event in the preceding seven calendar days
Default selection share	Selected-provider count divided by default probes in the provider-model-day cell
Rate-limit share	Router-published rate-limited count divided by successful plus rate-limited counts in the published window
Author-operated endpoint	Provider name matches the model author under the checked author-provider crosswalk
Direct parity	Direct API quote equals the contemporaneous router quote within machine tolerance

Table 5: Key variable definitions.

The capture scheduler targets five-minute cadence, but network and source failures create irregular intervals. All duration variables use actual UTC timestamps. Missing captures are never forward-filled for event timing. A quote that is unchanged across a gap contributes exposure but cannot locate an unobserved intermediate round trip; historical sources are therefore used as a lifecycle sensitivity, not to impute high-frequency paths.

The platform’s five-minute and thirty-minute counters can overlap mechanically. Regressions use one horizon at a time. Capacity ceilings are displayed metadata, not contractual commitments, and missing ceilings are not coded as infinite. Derank transitions and rate-limit onsets are router enforcement events; they do not reveal which user’s request caused the event.

B Independent grid-null algorithm

For each observed model-day (m, d) with $N_{md} \geq 2$:

1. Compute the observed median log price $\tilde{p}_{md}$ and provider deviations $u_{imd} = \log p_{imd} - \tilde{p}_{md}$.
2. Pool deviations over model-days after the same endpoint and variant filters used by the observed statistic.
3. Draw N_{md} deviations independently with replacement, add $\tilde{p}_{md}$, exponentiate, and snap to the declared grid.
4. Record whether the two smallest simulated prices are equal. Aggregate over model-days and repeat the full simulation.

The primary grid is one cent per million tokens; the sensitivity is ten cents. The simulation preserves the distribution of market thickness and model-day price levels but removes cross-provider dependence. Because the deviation pool is empirical, it does not assume a parametric cost distribution. Its main limitation is that pooled deviations combine provider heterogeneity and may retain atoms from the observed coordination process.

C Randomized-crossover protocol

Each eligible block chooses a hot model with at least three publicly quoted providers. It freezes the contemporaneous provider set and computes cheapest, second-cheapest, and random pinned candidates. A deterministic pseudorandom seed permutes the four policies and the model order. The assignment row is written before requests begin. Requests use a fixed minimal prompt and one output token.

The primary outcomes are HTTP-level success and a valid generation identifier. Secondary outcomes are latency, observed spend, selected provider, and fallback attempts when exposed. Missing cost is not recoded as zero. A block is valid only if assignment replay passes, the quote snapshot predates the request, and each declared policy is attempted exactly once. The confirmatory prefix stops at 500 valid attempts per pinned arm; later rows are a separate continuation sample.

The three primary hypotheses compare default success with each pinned arm. The family uses Holm adjustment. Randomization inference permutes policy labels according to the original assignment mechanism within blocks. The first-position estimator is primary for arbitrary carryover. Pooled position-adjusted contrasts and model-policy heterogeneity are secondary. A flat cheapest/second/random gradient is reported as evidence against rank-specific stale-quote refusal, not as equivalence unless the registered equivalence margin is met.

D Steering-audit robustness

The unrestricted audit includes every provider-model-day that is currently cheapest on the public surface and has at least one default probe. Model-day fixed effects remove common demand and quote-level shocks; price-rank controls compare providers at the same current prominence. Standard errors cluster at provider and model-day in the promotion specification.

Three robustness layers are preregistered: (i) require at least one successful pinned request for the provider-model pair; (ii) use only regions and prompt lengths for which all compared providers pass eligibility; and (iii) instrument recent-cut visibility with randomized quote-surface exposure if the platform permits a safe visibility experiment. Current data support the direction in layer (i) but not its sample size. No visibility instrument has fired.

E Transparent-compute comparator

A separately preregistered Akash study links public multi-provider bid sets, the accepted lease, and exact on-chain termination reasons. The launch backfill replayed 2,929 retained closed leases with a 100% exact-ID match. Its pre-preregistration support set contains 36 linked multi-provider choices: 20 accepted prices above the lowest bid and 16 at the lowest. All 36 have the fixed 300-block follow-up, but only 20 close in that window; none records a provider, escrow, or non-owner close. This is zero-event support, not evidence of no effect.

The prospective cohort uses a second-snapshot inception rule, masks confirmatory outcomes, and has a fixed release of 2026-08-15. The comparator is excluded from the paper’s three facts because its market object is a compute lease rather than a token-priced inference request. Its role is methodological: it shows what becomes identifiable when bids, acceptance, and termination are public.

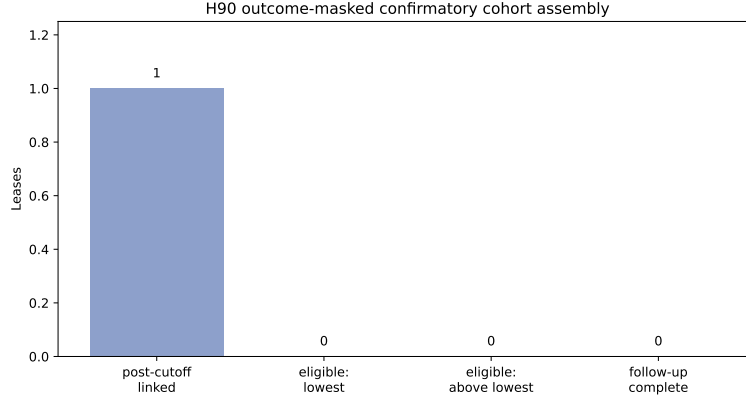


Figure 2: Prospective Akash cohort assembly at launch. The absence of eligible post-cutoff observations is a support diagnostic; outcomes remain masked until the fixed release.

F Claim ledger and promotion gates

Claim	Current evidence	Missing identification	Promotion gate
Administered menus	Frequency, size, grid, clock, day-split gap AUC	Long-run frailty and strategic response	30 panel days, both vintages
Author focal anchor	Grid null and 65/72 author-price matches	Exogenous author move	First qualifying author repricing
Manufactured firmness	Randomized level, flat rank gradient	Full randomization precision and transport	500 attempts per pinned arm
Negative steering memory	Rank-conditioned audit, same-sign eligibility subset	Complete eligibility or visibility instrument	Eligibility-complete panel
Brown-MacKay mechanism	Cadence premium	Slow-initiator/fast-responder exposure	30 risk pairs after frozen cadence
Retry externality	Forward/backward asymmetry, small IV estimate	Cross-provider retry and user value	Incident instrument plus attempt logs
Literal front-running	None	Cross-user order and quote-update sequence	Negotiated router/provider event logs
Global welfare loss	None	Values, quality, costs, transfers, capacity response	Structural measurement contract

Table 6: The ledger distinguishes a supported market fact from the stronger mechanism or welfare claim it may motivate.

G Reproducibility

All tables are generated from versioned Parquet and JSON artifacts. Capture rows carry source timestamps, retrieval timestamps, and source-run identifiers. Probe assignments include a study identifier, block identifier, seed, intended policy, position, and quote-snapshot key. Prompt payloads and secrets are excluded. The repository contains the analyzers, preregistrations, claim-boundary documents, and machine-readable summaries. The four-panel figure reads only the frozen `paper_estimates.json`; it intentionally refuses the local legacy H80 summary because that file predates the randomized release. At each registered gate, the paper estimates, source manifest, and rendered PDF must be updated in the same commit.